

Strengthening Ukraine's Electric Grid Against Russian Cyber and Kinetic Attack

*Recommendations for Grid Hardening, Decentralized Topology,
Drone Capacity Building and Cyber Army Coordination*

Developed by

Tim Roxey

Eclectic Technologies

Critical Infrastructure Cybersecurity and AI Assessment

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Executive summary

Russia's sustained campaign against Ukraine's power system—combining targeted cyberattacks on industrial control systems with kinetic strikes on generation and transmission assets—is the most consequential assault on a national energy network since World War II. The lessons from this campaign reach far beyond Ukraine. Any NATO member with centralized generation, legacy control networks, and reliance on cross-border electricity flows faces a comparable threat. Alliance policy toward and assistance for Ukraine should therefore signal a readiness to act more broadly—especially for Baltic and Nordic states that are at heightened risk of similar coercion.

Section A — Hardening the technology network

This section examines measures to protect Ukraine's operational technology (OT) and communications infrastructure that control the grid. Building on recommendations provided to Ukrenergo leadership shortly before the February 2022 invasion, it proposes a strategy of dynamic network management: routinely changing network topology, IP addressing, and hardware configurations so that an adversary must repeatedly re-acquire the information needed for a successful cyber intrusion. These techniques increase the cost and complexity of an attack and improve defenders' ability to detect and isolate intrusions.

Section B — Protecting physical grid topology

This section focuses on the grid's physical layout. Ukraine's centralized generation model—few large plants feeding distant loads via long transmission corridors—creates concentrated, high-value targets for standoff weapons. A coordinated cyberattack against the four nuclear sites, for example, could trip all nuclear generation simultaneously and trigger cascading failures that might collapse the southeastern grid as far as Kyiv. The section outlines mitigation options to reduce single-point vulnerabilities and enhance resilience against both kinetic and cyber disruptions.

Section C — Weapons access and AI-enabled systems

Section C addresses two related challenges. First, allied reluctance to enforce closed airspace or to permit long-range strikes has allowed Russia to employ standoff munitions—missiles, glide bombs, and long-range drones launched from beyond Ukraine's reach—against civilian infrastructure with relative impunity. Building domestic weapons production capacity in Ukraine would reduce dependence on external supply chains and the operational restrictions that partners sometimes impose. Second, battlefield pressure is accelerating the integration of artificial intelligence into drone operations. AI-enabled drones already demonstrate markedly higher effectiveness—improving resistance to jamming, enhancing autonomous logistics, and increasing the success rates of reconnaissance and strike missions. These trends change the operational calculus for both defense and escalation.

Section D — The Cyber IT Army and legal/organizational challenges

This section examines Ukraine’s crowd-sourced Cyber IT Army and the challenges that arise when civilian volunteers undertake military cyber operations. Integrating volunteers raises legal, organizational, and coordination problems. Most importantly, it blurs the legal distinction between combatants and civilians—central to international humanitarian law—because non-combatants who carry out cyber operations can become legitimate targets and may lack clear legal protections or status. The section recommends frameworks for oversight, legal clarity, and safeguards to minimize the risk of retaliation and ensure accountability.

Background of Russia Attacks Against UA Grid

This chapter reviews the attacks on Ukraine and highlights key findings and trends, emphasizing their significance to the U.S. and EU energy grids. Following Russia's invasion on February 24, 2022, the focus has shifted to the eastern regions of Ukraine, targeting critical infrastructure, especially the electrical grid.

This paper addresses a central question: What preliminary insights can we gain into the important differences in thinking, approach, and character of infrastructure, and what might the U.S. military be called upon to do, especially in helping ally and partner countries sustain their critical systems under attack? Given current geopolitical realities, it is difficult to determine what actions the US Military should take to support the Western European Allies. However, as relations normalize, the actions described in Sections A through D would be most useful for supporting the US Allies. As always, the US Allies should consider these actions independently.

Hardening Ukraine’s Electric Grid Control Network

The Technology Network and Its Role in Grid Defense

Based on the author's work spanning several years and careful monitoring of open-source reporting from 2018 to 2021, it was believed that Russia intended to invade Ukraine.¹ Furthermore, a computer network attack would likely precede the invasion; given this, the Ukrainian Technology Network's physical and logical infrastructure needed to be modified to disrupt Russia’s plans.

This section analyzes preliminary messaging to Ukrainian grid operators developed by a team of grid specialists to help prepare Ukraine for an anticipated ground invasion. This message was sent on February 6th, just before the ground invasion on February 24th, 2022.² The analysis focuses on industrial control systems within the energy infrastructure, specifically the Technology Network. The recommended actions are broken into three parts: 1) do immediately, 2) do as soon as possible, and 3) do if you still own the resources. Since the invasion was considered imminent, the messaging emphasized “do immediately.”

¹ Based on the author’s private sector support of Ukraine following the 2015 and 2016 attacks against their grid. Additionally, the author worked with the Ukrainian Ukrenergo personnel during late 2021 and early 2022 to help them adjust their future state design plans. This was a private work effort by the Author's distributed team.

² The team that developed the February 6th letter was called the Hydra Team. The Hydra comprises individual subject matter experts with decades of experience in critical infrastructure. The members of this team act independently of the companies they may work for and agree, in advance, to share their expertise with the Hydra Team.

The Technology Network³ is the backbone of the industrial control systems (ICS), and Operation Technology (OT) networks used to control Ukraine's electric grid infrastructure. The other side of the network is the Office Network, which handles the office and desktop environments and is where most employees interact in their day-to-day work. There are very few connections between the Office and the Technology Networks, all of which have firewalls in place. Hardening the Technology Network is a key element of network defense, as it helps deter attacks on control systems. The attackers' frustration stems from the requirement that the adversary enumerate the targeted Technology Network before each attack.

Assumptions for Hardening the Technology Network

The Hydra Team developed assumptions based on informal conversations, as outlined in a January letter to the Ukrenergo leadership.⁴ Each recommended action was designed to reinforce or challenge one or more of these assumptions. As working assumptions, the Hydra Team provided no evidence of their validity; instead, they were based on OSINT (Open-Source Intelligence) and the team's understanding of adversarial capabilities and capacities. Furthermore, these assumptions were informed by insights into adversary behavior regarding Tools, Techniques, and Procedures (TTPs) drawn from decades of experience among Hydra Team members.

The December 2015 and 2016 attacks demonstrated Russia's deep access to the Technology Network, and the Hydra Team assessed that Russia possessed a comprehensive network map of all relevant components. The 2016 attack was particularly revealing: modular malware targeted Ukrainian electricity infrastructure, significantly disrupting grid operations. Analysis showed that the CRASHOVERRIDE malware framework could be customized to execute multiple actions on industrial control assets, including manipulating digital relays. Cybersecurity firm Dragos Inc.⁵ and ESET⁶ reported on this malware. These reports suggest Russia possesses considerable knowledge of the Ukrainian Office and Technology Networks.

Given this level of demonstrated access, the Hydra Team assumed that Russia maintained a complete topology map of the Office and Technology networks and held access to most critical components.⁷ Changing network details just before an invasion would force Russia to "rediscover" the details to support its computer network attacks. Periodically changing these details would force Russia to continually scan the Technology Network to support its ongoing cyberattacks.

Russia likely followed the steps outlined in the SANS ICS Kill Chain to gather sufficient information to plan and execute attacks against the Office and Technology Networks.⁸ The first part of this Kill Chain involves Russia regularly conducting reconnaissance to gather details

³ The Ukrainian Technology Network (it is named this in system documentation) comprises operational technology (OT) and information and communication technology (ICT) devices that monitor and control the production and distribution of electricity.

⁴ The January letter was delivered to the US Embassy and the technical leads for the Ukrenergo on February 6, 2022. The content was in development since late August 2021. It is called the February 6th letter since that was its delivery date.

⁵ A full forensic teardown is available at: <https://dragos.com/blog/crashoverride/CrashOverride-01.pdf>

⁶ A link to the ESET press release is at: <https://www.eset.com/us/about/newsroom/press-releases/eset-discovers-dangerous-malware-designed-to-disrupt-industrial-control-systems/>

⁷ Historically, Ukraine belonged to the Soviet Union, and most of its critical infrastructure facilities were built during this occupation. Hence, Russia has detailed network maps of all Office and Technology networks.

⁸ The Cyber Kill Chain is a process that adversaries follow to ensure a successful attack against an ICS device. Further information is at <https://www.sans.org/white-papers/36297/>

about the Office and Technology Networks. Using tools like Shodan⁹, Russia can scan the IP space of the electric grid control network to identify any Internet-connected ICS devices. Each discovered ICS device represents a potential entry point for a Russian cyberattack. It is assumed that Russia has identified all the devices exposed to the Office and Technology networks. Following the reconnaissance step in the Kill Chain, the weaponization phase involves developing specific, tailored attacks against each device discovered during reconnaissance. These tailored attacks, developed during this weaponization phase, would likely be highly effective.

The prepared attacks against the Office and Technology Networks are now ready to execute whenever the adversary's commander chooses to use them. To keep their prepared attacks ready, Russia would need to maintain up-to-date information on targeted networks, which would drive several recommendations to promptly change the Technology Network addressing schemes.

Summary of Assumptions

1. RU has gained pervasive access to the UA electrical systems Oblenergo and Ukrenergo IT network systems

It was repeatedly shown by RU attacks in December 2015 and 2016, among many other examples.

2. RU has a complete electrical system one-line diagram showing its interconnects to ENTSO

At a minimum, <https://www.entsoe.eu/data/map/>

3. RU has used SHODAN, or something similar, to explore the IT Network IP space for each large utility and has enumerated all exposed IT systems/devices.

Note: Importantly, this enumeration is likely to have occurred across all other critical infrastructure sectors.

4. Russian sympathizers working within the various Ukrenergo facilities involved in their construction and operation would allow even non-public information to be available to the Russian attack planners.

5. RU has developed an attack package for each significant device identified, including network mapping to the target device

Note: This assumption drives IP address changes, such as changing street signs to confuse enemy troops.

6. RU Adversary Access Teams have developed credentials for the OT networks of interest, including SCADA credentials.

Note: In the UA attack in December 2015, the RU was inside the network for at least 6 months before the attack. This persistence implies that Russia adjusts the access credentials as needed.

⁹ Shodan is one of many network enumeration tools designed to detect industrial control devices connected to the Internet. The ICS devices on the Technology Network, typically not connected to the Internet, may have a connection through oversight. Typically, the oversight is due to the connection not being removed after installation or maintenance. [https://en.wikipedia.org/wiki/Shodan_\(website\)](https://en.wikipedia.org/wiki/Shodan_(website))

Recommendations for the Technology Network based on the assumptions.

The recommendations below are based on the Ukrenergo network design presented to the author during a 2017 visit. At that time, the Ukrenergo was building a future-state network design. The CEO of Ukrenergo requested an assessment of potential design weaknesses that adversaries could exploit. People from a US National Lab and several of the author's Hydra Team members reviewed this network design and provided specific guidance. It is assumed that the final "as installed" network reflected these suggestions and was in place before the Russian invasion.

The Hydra Team's first principal assumption, which led to these recommendations, was that Russia would launch an invasion very soon. This sense that Russia was preparing prompted the author to convene a Hydra Team conference call, where he laid out his concerns.

Do now – This applies to the Technology Network circuits.

Focus on the output breakers of nuclear generating stations and their SCADA circuits. Opening transmission breakers close to the plant's Generator Step-up transformer while the station is at full power and synchronized to the grid will cause the nuclear power plant to trip. The associated units will immediately trip upon opening, requiring as few as six circuit breakers at the Zaporizhzhia Nuclear plant, removing 5,700MW of generation. The adjacent seven fossil units will not be able to ride through the 5,700MW loss; they will also trip, removing an additional 3,650MW. This combined loss of 9,350MW will likely cause a system-wide disturbance, shedding generation across most of the Ukrainian grid.¹⁰

Once Russia decides to launch the impending attacks, it is assumed that they will use a cyber attack against the four nuclear plant sites in Ukraine. Such an attack would likely trip all of Ukraine's nuclear power plants, which they could reach simultaneously. This is achieved by opening only one breaker close to the generator step-up transformer adjacent to each Nuke Plant. The sudden loss of all Nuclear Power Plants will trigger a cascade that will trip additional generation. It is unclear how far the cascade will extend, but it will likely shut the entire Southeastern grid up to Kyiv.

At all other significant generation stations, prioritize generation units by size, critical loads, and voltage-support capability.

- 1) At all critical Generation and transmission switching stations, replace some protective relays from stock and reconfigure (include firmware from an older vintage)
- 2) At all nuclear power plants, block remote access to all substations
 - a) Place all generator output relays into local control mode
- 3) In the Technology Network of critical generation plants, change some key routers/switches
- 4) Put new Rugged Coms in front of your substation comms
 - a) This disrupts covert, persistent access by the bad guys you don't know about...
- 5) Stop all Web Server accesses to the Vendor A RTUs –
 - a) any access to these RTUs and their web server allows anyone to gain user credentials and change configuration – no fix other than to restrict access

Do Soon – This applies to both the Office and Technology Networks.

- 6) Locate the restoration plan for the business and power control center servers. Retrieve the "Gold Copies" of the servers and ensure they are readily available

¹⁰ This type of attack against nuclear facilities is called the Queen's move. The focus on tripping nuclear plants is because these plants cannot be recovered from a trip as fast as a fossil-fueled plant can. Typically, the nuclear plants will require up to 72 hours before restart.

- a) These should already be done. Find them and keep them convenient to hand
- 7) Consider making the primary and backup Control Network IP addresses less static
 - a) Change the IP structure by forcing all nodes to re-establish their IP address on a non-predictable basis.
- 8) For all Technology Network
 - a) Have a plan available to disable MGWs (Media Gateway servers) that handle Voice, Data, and Fax
 - b) The MGW services come in under the firewall protection; hence, the FW routers cannot filter out the adversary traffic
- 9) For All Office Network
 - a) Have a plan to disable both the MGW and IP-PBX (IP-based phone) services
 - b) Both the MGW and the IP-PBX come in under firewall protection, and hence, the FW cannot filter out adversary traffic
- 10) Disable the Wireless Protected Network in the Office network level, General Office. The WPN typically comes into the office Switch, which is already inside the firewall.

Consider – As time permits

- 11) Extend the relay replacement program (item 1) to remaining non-nuclear generation and lower-priority transmission switching stations; reconfigure with firmware from an older vintage where feasible
- 12) Cut off remote access back to the vendor for the Security Information and Event Management (SIEM) System
 - a) If you have an older version (yes, an older version) of the SIEM, revert to it and use it.
- 13) Limit all ICCP traffic links between centers –
 - a) If you need data, load sharing/dispatch, etc. – use the fax and telephone
- 14) Cut off all OSI Pi links and all RCM (maintenance links)

The tactical recommendations above address the immediate threat. Beyond these discrete actions, a broader strategic shift in how the Technology Network is managed would compound its defensive value.

The recommendations for the dynamic network management scheme address the Technology Network, which defines the network topology. As called for in the February 6th letter, this network's topology should be changed periodically, requiring the Russians to expend additional resources to reacquire the information needed to conduct a computer network attack. One approach to making wholesale, large-scale changes to the Technology Networks topology is to use software-defined networking (SDN) and a mix of ICS vendor products. Periodic topology changes and the mix of ICS vendors will make network exploitation more difficult by requiring the adversary to repeatedly enumerate the entire Technology Network. It should be noted that the ideal periodicity for SDN to force a topology change should be random, not on a predictable schedule.

The downside of this suggestion is that, when applied on an irregular schedule, making large-scale topology changes to the Technology Network increases management effort. This is an additional burden on a stressed Ukrainian network engineer's day. Still, it offers the advantage of confounding the enemy's ability to monitor the evolving OT network. This technique would increase the adversary's cost of exploiting this changing network. Shifting to a software-defined network and more advanced, secure networking technologies will increase the cost of any computer network attack for Russia.

Remaining Task

- Assess the possible move of the present Ukrainian Technology Network to a software-defined networking topology.
- Move to a diverse technology blend of hardware and software suppliers in the Technology Network. It must be assumed that adversaries will map the entire IP architecture of the Technology Network, and that hardware-level equipment swaps will increase their costs.
- Inform the Nordic, Baltic, and European States of this Ukrainian methodology and encourage them to replicate appropriate parts. Although more difficult to manage, this concept of using a dynamic network addressing scheme applies directly to other nations of interest.
- Using US military services to support this network management is a reasonable additional burden for the US military.

Note: Given today's evolving political dynamics, NATO and EU member states should seek support from private-sector companies such as ESET.

B Decentralized Generation Grid Topology

Ukraine's electric grid operates under a centralized generation model, in which a relatively small number of large plants serve distant loads via long transmission corridors. These power lines and generation sources are susceptible to kinetic attacks from Russian missiles, drones, and conventional forces. The collocated Technology Networks that control these systems are equally vulnerable to cyberattack. The result is a set of concentrated, high-value targets—and Russia has demonstrated a willingness to strike them.

Since the beginning of the war, Russia has repeatedly targeted civilian energy infrastructure—actions that, a short time ago, would not have been believed possible in a European conflict. The bombing of power plants, the occupation of nuclear facilities, and the systematic destruction of grid components all follow a pattern of coercion through infrastructure denial.

Western allies have been slow to anticipate and respond to this pattern of escalation. The implications for allied grid resilience—particularly for the Baltic and Nordic states—are addressed further in Section C. The immediate challenge for Ukraine is rebuilding a grid that continues to be destroyed.

Replacements are required as destruction continues.

The Ukrainians are heroic in maintaining their infrastructure, but they need support. With winter approaching, Russian attacks are targeting areas of Ukraine's infrastructure that keep the population fed, housed, and warm, specifically agricultural distribution, civic centers, and energy infrastructure.

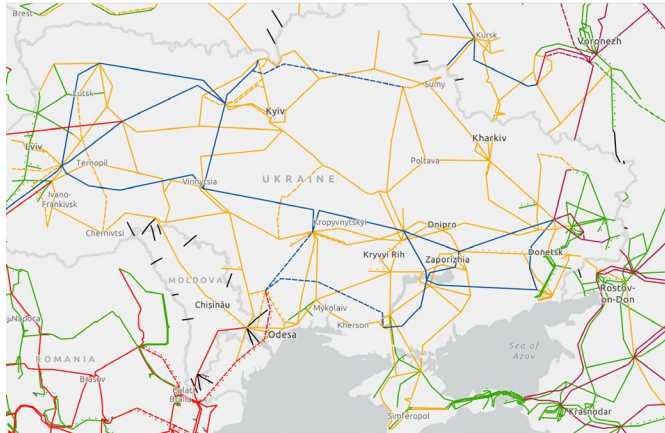


Figure 1- Ukraine Grid

The Ukraine Grid diagram illustrates the various transmission lines and their distribution from larger generation facilities. Blue represents 750 kV lines connecting four nuclear power plants: Rivne, Khmelnytskyi, South Ukraine, and Zaporizhzhia. Orange denotes 330 kV, red indicates 380-400 kV, and green signifies 220 kV. The interactive e-NTSO maps also highlight smaller thermal plants and lower-voltage transmission corridors. These open-source maps demonstrate that the major facilities and transmission lines are publicly accessible.

International Support and Equipment Replacement

Following the initial grid attacks in 2022, Ukraine replaced destroyed grid components with available spares and scavenged equipment from disused facilities. The intent was to restore the grid's damaged components as quickly as possible. As elements of the grid continue to be destroyed during replacement equipment installation, Ukraine has run short of, or even out of, the necessary spares.

Ukraine contacted the international community to secure replacement components to keep the as-built centralized generation grid operational, with a focus on repeatedly damaged substation transformers. In 2023, a large, multinational, public-private support program provided replacement materials for damaged or destroyed energy infrastructure. Multiple governments delivered these replacements to Ukraine. Subsequent Russian attacks destroyed some of these replacements with ease because their exact locations were already known.

The material in North American electric utility warehouses could be very useful to Ukraine as it struggles to keep its electrical infrastructure operational this winter. Some of this inventory had been in warehouse storage for some time, with utilities covering the warehousing carrying costs. Upon receiving the request from Ukraine, US utilities determined they did not have an emergent planned use for the materials during the winter season. Still, they were carrying them at an inventory control level. These carrying costs are not trivial and cover all necessary maintenance for large equipment, including during warehouse storage.

After a brief coordination effort by North American utilities, they decided to use their existing warehoused inventories to fulfill some of Ukraine's requests. The utility then replenished its donated equipment with vendor-supplied items. The utility's risk in this donation scheme is that it might need the materials before receiving replenishment orders, although many consider this risk very low. The utility would benefit from this donation process by refreshing its inventory, which is often welcome. Although the Ukrainian grid is relatively small compared to the larger

North American grid, the availability of spare parts from NA inventories soon diminished. The inventory refresh for the donating utilities signals new equipment rolling out of supplier plants, which would pressure the suppliers to increase production. This supply-chain ripple effect allowed the international community to share some of the burden that Ukraine was feeling from the Russian attacks on its grid.

Recent attacks and replacement efforts have emphasized Ukraine's centralized grid's vulnerability to Russian assaults. To address this, a swift transition to a decentralized model is essential. Planning each replacement requires determining the load paths linking critical facilities to their dependent generation resources. Identifying these crucial load paths closely resembles the concept of the “Cranking Path,” which is vital for system restoration after a collapse.¹¹ Once these paths are identified, they can be prioritized. When a generation resource is destroyed and must be replaced, the new generation source should be as close to the critical loads as transmission systems permit. By employing this strategy and having the various Critical Load Paths already prioritized, a gradual transition to a decentralized grid topology is achievable. Although this process may take several years, prioritizing load path replacements can begin at any time.

Note: It's acknowledged that modifying the topological grid is very challenging for Ukraine while the grid is operational and under attack by Russia. Nevertheless, such a change is strongly recommended for the Nordic and Baltic states. As the energy sector appears to be a primary target of Russian coercion, any actions these states can take to enhance the robustness of their grids should be considered.

A path forward.

One of the many challenges exposed by the war is that the centralized grid model is vulnerable to kinetic attacks. A potential solution is to transition the centralized grid to a decentralized model. While this may seem straightforward, the reality is much more complex. Transitioning from a centralized generation topology to a more resilient decentralized model in a nation at peace would be difficult; however, shifting to a decentralized grid during wartime is exceedingly challenging.

Shifting to a decentralized generation topology is important for strengthening the reliability of energy support for defense-critical energy infrastructure (DCEI) and civilian-critical energy infrastructure (CCEI).¹² This more decentralized topology would require more standoff weapons to achieve the same megawatts lost per weapon used, considerably reducing the MW Lost/#weapons ratio. This metric—megawatts lost per deployed weapon—provides a quantifiable basis for prioritizing which load paths to decentralize first, and the reduction in this ratio directly increases Russia's cost of attacking the grid. Grid operators should explore additional ways to reduce this metric; one such approach is described below.

Consider the DCEI and CCEI aspects of the largest generation sources, such as the 6 GW Zaporizhzhia nuclear power plant (six NPP) and the adjacent 2.8 GW fossil-fueled thermal

¹¹ A cranking path links an active generator (often a black start unit) to another generator needing electrical power for startup. Its purpose is to provide the essential startup power to inactive generators, allowing them to be started. In this context, a Critical Load Path links a Defence or Civilian critical facility and the primary generation source providing their power.

¹² DCEI Defense Critical Electric Infrastructure is any electric infrastructure that serves a Critical Defense Facility. In like manner CCEI is any electric infrastructure that supports any critical civilian infrastructure. CCEI includes elements such as mass evacuation centers, hospitals, large commercial complexes, and city centers.

power plant (seven TPP). These generation facilities should be analyzed to determine the DCEI and CCEI loads they support. These two plant sites, with 13 units in total, account for almost 50% of Ukraine's energy needs during the winter peak.¹³ All 13 of these units are presently under Russian control.

Once the DCEI and CCEI loads are determined, replacement generation should be positioned closer to these loads rather than in the locations that were just destroyed. For DCEI loads, it is essential to prioritize the incremental replacement of lost central generation resources with distributed generation and microgrid development at these load centers. This approach can be planned and executed using remote resources provided by a coalition of NATO and EU partners. Most importantly, this analysis process and the redeployment of replacements after the destruction of the originals should not rely on the already burdened Ukrainian resources but instead be supported by allies. Establishing teams to carry out these tasks will become a key capability for the Nordic and Baltic states should they need similar efforts, hopefully before any destruction occurs.

Locating smaller generators near essential facilities reduces the risk of large-scale disruptions from attacks on a few major generators and high-voltage power lines. Any planning for replacements should aim to minimize the metric of megawatts lost per deployed weapon, expressed as (MW lost / # weapons). The attrition rate of incoming glide bombs and missiles from Ukrainian air defense can be incorporated into this calculation. Evaluating the centralized generation grid strategy with this metric reveals which DCEI or CCEI load paths should be prioritized for decentralization.

Note: This represents the primary wartime application for microgrids. Small generators deliver essential electrical services to commercial or government facilities.

Making it happen.

The material and logistical support from the private sector and military planners would be vital to implementing this strategy. Using professional strategic planners would be useful in obtaining the necessary funding from organizations such as the World Bank¹⁴, NATO¹⁵, and the European Union¹⁶. As shown in the footnote links, each of these three organizations has already studied how to secure funding for reconstructing Ukraine's critical infrastructure. Additional funding sources are scattered globally and include frozen Russian reserves.¹⁷ Although the US has suggested using the entire seized funding, the Western Alliance supports using only the interest earned on the total, not the principal.

¹³ Kyiv Post article by Leo Chiu, Olena Hrazhdan | July 6, 2024, <https://www.kyivpost.com/post/35398>

¹⁴ Potential funding source for the reconstruction of critical infrastructure destroyed in war. https://en.wikipedia.org/wiki/World_Bank_Group

¹⁵ This Atlantic article covers the details of the NATO Stabilization and Reconstruction efforts (NATO S&R).

https://www.atlanticcouncil.org/wp-content/uploads/2006/09/061021-How_Should_NATO_Handle_SR_Operations_0.pdf

¹⁶ The EUs ability to fund the reconstruction effort is shown in this example. https://eu-solidarity-ukraine.ec.europa.eu/eu-assistance-ukraine/recovery-and-reconstruction-ukraine_en

¹⁷ It is estimated that upwards to 300 billion USD are being held by different sovereign nations. The US has proposed that these funds be seized and utilized to support the rebuilding of Ukraine. Each sovereign nation has their own set of issues with this UA+S proposal. <https://www.reuters.com/world/europe/what-where-are-russias-300-billion-reserves-frozen-west-2023-12-28/>

The U.S. military can support civilian efforts through NATO's Partnership for Peace Program, which would coordinate all signatories to the agreement.¹⁸ Using the military to support civil authorities in developing and implementing a plan to transition Ukraine's electrical grid is a major effort. It would be a roadmap for other countries in the region to do likewise. Performing the change to a decentralized grid during equipment replacement following Russia's destruction would mitigate the impacts of such a grid migration. This replacement-based restructuring approach should be recommended to the Baltic and Nordic states, which face similar threats from Russia. These states could implement these changes before any potential destruction, thereby improving the resilience of their electric infrastructure in peacetime and in preparation for war.

Implementing plans includes replacing damaged or destroyed generation resources and moving replacement generation assets closer to the DCEI and CCEI. Developing local or edge generation near critical infrastructure requires significant planning. This effort can benefit from studies performed in the US, as discussed in the Defense Science Board's "The More Fight, Less Fuel" policy document.¹⁹ The magnitude of resources required to implement the modifications will require support from the private sector, EU countries, the US, and its NATO Allies.

Remaining Task

- Plan the gradual transition from a centralized to a decentralized structure based on a replacement strategy.
- Establish the necessary military logistical support groups to develop the required planning.
- Prepare the required DCEI and CCEI list and prioritize
 - Establish the Critical Load Paths for the prioritized DCEI and CCEI facilities.
 - Develop work packages for the top ten DCEI and CCEI Load Paths that address the equipment needs and logistics for the planned modifications to the generation sources and load paths. This work mirrors the approach used by various energy companies to build new energy networks.

C Hesitancy, Drones, Capacity Building

Allies' Hesitancy

The Russian use of standoff technologies, including missiles, glide bombs, and remotely operated drones, disrupts the traditional military defense doctrine employed in troop deployments. Drones operated locally by both Russia and Ukraine have altered the dynamics of conflict at the frontlines. Ukraine's defense against Russian drones poses significant challenges. Missiles and glide bombs fired from great distances, and the absence of closed airspace, give Russia a difficult-to-overcome advantage. These missiles and glide bombs highlight Ukraine's struggle to defend itself effectively.

Recently, Russia has avoided using its aircraft in areas very close to the front lines, opting instead to stay back and launch missiles and glide bombs. This strategy compels Ukrainian

¹⁸ Partnership for Peace (PfP) is a program that facilitates bilateral cooperation between NATO and individual Euro-Atlantic partner nations. It enables these partners to establish a tailored relationship with NATO by selecting their areas of focus for collaboration. https://www.nato.int/cps/en/natohq/topics_50349.htm

¹⁹ Defense Science Boards' More Fight Less Fuel. <https://apps.dtic.mil/sti/tr/pdf/ADA477619.pdf>

defenders to engage smaller targets rather than the larger aircraft from which the standoff weapons originate. Ukraine's inability to reach the sites where these standoff weapons are launched, coupled with a lack of robust military support from its allies to disrupt these systems, has been debilitating. The hesitation of the Allies to enforce closed airspace for Ukraine has directly enabled Russia to continue its ongoing assault on Ukrainian civil infrastructure.

There is a crucial opportunity to help counter Russian standoff weapons. Establishing the required infrastructure for weapon system production in Ukraine would eliminate dependence on external supplies and the restrictions imposed by allies on their use. This strategy would allow Ukraine to boost its capacity to create countermeasures against standoff weapons and target military installations deeper within Russia. The limitations imposed by allies significantly hinder the execution of Ukraine's defense strategy. Initiatives to optimize supply chains should be integrated with efforts to build internal capabilities and expand capacity.

Drones

Ukraine's use of both aerial and sea-based drones is advancing its role in modern warfare. The drones employed in the Ukraine conflict perform many functions humans typically manage. Ukraine is utilizing drones in several novel and significant ways during the ongoing conflict. However, effectively integrating drones into the overall strategy and day-to-day field tactics is challenging.

One recent development among many that Ukraine has been compelled to pursue since Russia's invasion is the adoption of artificial intelligence-enabled drones. In each drone application area detailed below, the use cases for AI-powered drones are still being defined. A recent Kyiv Independent article reported that the use of AI in Ukrainian drones has increased the drone's "Kill Rate" to as high as 80%.²⁰ Additionally, in a recent article for the Henry Jackson Society, David Kirichenko highlighted the increasingly significant role that artificial intelligence (AI) plays on the battlefield.²¹ These improvements are evident in a drone's ability to counter Russia's signal jamming and in its targeting capability.

Note: The use of AI in these applications is very new. Its use can only improve with further battlefield experience. It is difficult to overstate the importance of supporting Ukraine's domestic technology development efforts. Many other Western European states will likely need these types of drone improvements should the Russian war continue to escalate.

The following is a brief overview of drone use in Ukraine.

Surveillance and Reconnaissance:

Although this is merely a survey of recent drone usage, it is important to note that the war has pressured drone users to innovate. As recently as February 8, 2025, Tim Mak, Oleksandr Matviienko, and Mariana Lastovyria reported on an all-drone attack in the Counteroffensive Substack.²² The drone-only attack delivered strong results, with about 100 people affected, from

²⁰ The Kyiv article is found @ <https://www.kyivpost.com/post/40500>

²¹ The full article from David is found @ <https://henryjacksonsociety.org/publications/military-lessons-for-nato-from-the-russia-ukraine-war-preparing-for-the-wars-of-tomorrow/>

²² Link to Substack: <https://www.counteroffensive.news/p/the-first-ever-all-drone-assault?r=7jebo&triedRedirect=true>

logistics to operators. The ground troops were able to take and hold the Russian position after the drone attack.

Intelligence gathering:

Drones collect intelligence on enemy positions, troop movements, and logistical routes. This information aids in planning military operations and enhancing situational awareness. Matt Burgess writes in the May 6, 2022, issue of Wired, quoting Valerii Iakovenko, the founder of the Ukrainian drone company DroneUA, “Drones have changed the way the war was supposed to unfold.” “It is all about intelligence, gathering and transmitting data regarding enemy troop movements or positions, and correcting artillery fire.”²³

Targeting Artillery:

Armed drones can identify and target enemy artillery, as well as other military assets. This capability enables Ukrainian forces to strike efficiently while minimizing collateral damage. Initially, first-person view (FPV) drones were used for basic surveillance. As operators’ techniques advanced and the drones became more sophisticated, incorporating explosive packages became possible. Consequently, missions transitioned from simple surveillance to targeting artillery and personnel.²⁴

Cargo Transport:

Drones deliver supplies, including ammunition and medical equipment, to frontline units, helping maintain the flow of resources without putting personnel at risk. According to a Jane's article (behind a paywall), Simon Muderack, the CEO of Windracers and Distributed Avionics, disclosed that its UAV has been conducting intelligence, surveillance, and reconnaissance (ISR) while supporting resupply for the Armed Forces of Ukraine since 2023²⁵. These Windracers drones come equipped with an Autonomous mode, allowing them to deliver up to 100 kg of material to a location within 12 hours of flight time from launch.

Combat Operations:

Some drones are equipped with munitions to strike enemy forces and installations. This asymmetrical warfare tactic enables effective engagement from a distance.

As previously noted, drones are transforming the battlefield in modern warfare. They are redefining the concept of a frontline, and as this changes, drones will also alter many other combat operations. FPV drones typically operate several kilometers from the front line. This distance protects drone operators from conflicts at the front line.²⁶

Capability and Capacity Development

As stated, efforts to enhance Ukraine's domestic drone production capabilities and associated networking systems are crucial. In addition to strengthening Ukraine’s domestic capabilities, significant emphasis should be placed on technology transfer from EU member states to Ukraine, particularly in defensive signals technology to help suppress Russian attacks. The ability to

²³ Article from Wired. <https://www.wired.com/story/drones-russia-ukraine-war/> discusses the impacts of consumer-level drones being used by Ukraine.

²⁴ This March 26, 2022, article by Mariano Zafra, Max Hunder, Anurag Rao, and Sudev Kiyada discusses small drones' surveillance and search-and-rescue effectiveness. <https://www.reuters.com/graphics/UKRAINE-CRISIS/DRONES/dwpkeyjwkpml/>

²⁵ <https://www.janes.com/osint-insights/defence-news/air/uk-uav-maker-reveals-its-craft-in-use-by-ukrainian-armed-forces-since-2023>

²⁶ The “FPV drones in Ukraine are changing modern warfare” by Tomas Milasauskas, Liudvikas Jaškūnas. <https://www.atlanticcouncil.org/blogs/ukrainealert/fpv-drones-in-ukraine-are-changing-modern-warfare/>

develop resilient offensive drones, develop defenses against enemy drones, and manufacture these systems domestically is essential.

Once again, strengthening drone-system supply chains, including diversifying suppliers to reduce critical dependencies, is important.

Remaining tasks

- Develop strong NATO-based support for Ukraine's internal drone production
 - Western allies should work together to establish secure supply chains for the various materials that Ukraine's internal drone production requires.
 - Any efforts the Western allies exert in this regard will be useful in helping the allies produce their drones.
- Develop a strong policy allowing drones to be engaged by the citizenry
 - Much of Western allied countries' critical infrastructure is in the private sector's hands. As such, the security of the infrastructure rests first with the private sector, with federal authorities providing backup.
 - In some countries, such as the US, private companies are typically not permitted to take defensive actions against drones should they be surveilling or attacking their facilities. An established policy that allows the private sector to engage attacking drones provides a critical defensive capability.

D Cyber IT Army

Shortly after the Russian invasion of Ukraine, the Ukrainian military began forming an extensive cyber army of volunteers: A crowdsourced collection of individuals who wanted to support Ukrainian efforts called the Cyber IT Army. Initially, the participants underwent minimal vetting, but recently, the Ukraine IT Army launched a website to recruit new personnel (<https://itarmy.com.ua/?lang=en>). This site enables new members to join the initiative, provides a common platform for communication, and helps sort out roles and responsibilities. The newly established IT Army faced management challenges, including coordinating dispersed cyber operators with the centralized Ukrainian Army and protecting personnel within the IT Army.

A significant concern regarding volunteer hacktivist groups like the IT Army is their tendency to blur the boundaries between combatants and civilians, particularly when civilians engage in attacks. A Newsweek report²⁷ March 14, 2023, highlighted the challenges civilians face when undertaking standard military actions. Kubo Mačák, a member of the International Committee of the Red Cross, expressed concern about recent armed conflicts and the recruitment of civilian volunteers for military cyber operations. The distinction between combatants and civilians is a key principle of international humanitarian law (IHL). Civilians cannot be targeted, while combatants have legal protections—such as immunity from prosecution for killing enemy soldiers, provided they adhere to wartime laws. Mačák warned that if civilians participate in offensive cyber operations, such involvement could alter the legal implications. He noted that civilians who engage in the war effort may face risks, as they could lose the protection typically afforded to them under IHL. International humanitarian law shields civilians from attacks when they do not directly engage in hostilities, but if they do, they may be targeted during their

²⁷ <https://www.newsweek.com/ukraine-drafting-new-law-legalizing-volunteer-hacker-cyber-army-red-cross-1786814>

participation. The challenge is that the definition of "direct participation in hostilities" within cyberspace remains ambiguous.

"Even though not every form of civilian involvement on the digital battlefield qualifies as direct participation," Mačák said, "the danger is that it may be seen as such by the enemy, thus exposing numerous civilians to a grave risk of harm."

Using civilian volunteers in war, even online, undermines another important legal principle, accountability, Mačák said. "Civilians in such [volunteer] units do not fall under the military hierarchy; they do not fall under the military's disciplinary proceedings," he said. He said a clear Chain of Command is necessary to ensure accountability in wartime, establish responsibility for military actions, and punish violations of IHL.

By mid-2023, Ukraine was drafting a new law to legitimize this IT Army and safeguard its participants. By enacting a Ukraine-specific law based on the Estonian Cyber model, they can swiftly establish policies and legal frameworks to utilize the IT Army better.²⁸ The new law in Ukraine aims to define the roles and responsibilities of Army members and their command structure. It seeks to eliminate the IT Army's legal ambiguity and establish a clear reporting framework for the Ukrainian Army.

Resolving conflicts between the Cyber Army's operations and the Ukrainian Army's tactical engagements is an additional challenge, given the Cyber Army's unconventional command structure. It is difficult to align the various actions of the Cyber IT Army with the broader Ukrainian military strategy. There are significant opportunities to integrate the Cyber IT Army with the more experienced resources of NATO countries; however, NATO's cyber capabilities primarily focus on DDoS-style attacks rather than on missions to dismantle Russian assets. For operations intended to cause physical harm to Russian assets, a separate mechanism exists: NATO's Sovereign Cyber Effects Provided Voluntarily by the Allies (SCEPVA) framework, reaffirmed in 2022, allows individual member countries to conduct advanced cyber-kinetic operations on their own initiative rather than under NATO command. This SCEPVA framework allows member countries willing to undertake these types of advanced cyber-kinetic-specific missions²⁹ to engage on their own rather than with NATO³⁰. Even when NATO adopts this SCEPVA strategic concept, it does not guarantee that actions taken will align with Ukraine's battlefield strategy or the tactical situation on the ground. Nevertheless, the SCEPVA framework can alleviate some of the pressure on the currently overextended Ukrainian cyber engineers and enhance coordination among the decentralized actions of the Cyber IT Army and other strategic and tactical initiatives of Ukraine. The development of a new UA law, modeled on the Estonian Model, is a welcome control mechanism for the distributed IT Army and, at a minimum, establishes a chain of command.

The challenges of the Cyber IT Army are not merely organizational. They connect directly to the broader themes of this document: defending Ukraine's grid requires coordinated action across physical hardening, network defense, kinetic deterrence, and cyber operations. Without a legal

²⁸ The Estonian Cyber Defence League is a long-time collaboration of both public and private sector cyber security experts of Estonia, and emerged in the aftermath of the 2007 cyber attacks against Estonian information infrastructures, https://ccdcoe.org/uploads/2018/10/CDU_Analysis.pdf

²⁹ The SCEPVA program is found at <https://www.nato.int/docu/review/articles/2023/06/06/nato-and-strategic-competition-in-cyberspace/index.html>

³⁰ Of note, only France, the UK, and the US usually participate in SCEPVA actions. Again, the NATO members' cyber assets are typically only used for DDOS and other harassing campaigns.

and command framework that integrates volunteer cyber forces into this broader defensive posture, the risk of uncoordinated or counterproductive operations grows—and with it, the risk to the civilians who volunteer.

Remaining Tasks

- Convene a senior policy writers' group to draft an EU Cyber IT Army policy allowing citizens to support EU militaries in times of crisis.

Notes and Conclusions

Notes

Note: The central question driving this paper—what the U.S. military might be called upon to do in support of allied infrastructure resilience—is addressed throughout the four sections and synthesized in the conclusion below.

Given current geopolitical realities, it is difficult to determine what actions the US Military should take to support the Western European Allies. However, it can be suggested that, as relations normalize, the following actions would be most useful for supporting the US Allies. As always, however, the US Allies should consider these actions independently.

Conclusion

Attention must be directed towards which sections of this document the US military should adopt domestically. Collaborating with allies can clarify the steps required for domestic implementation. Although the US military has pinpointed several vulnerabilities on home turf, this information remains largely undisclosed, and our collaboration with allies has not reached its full potential. Analyzing experiences from the war in Ukraine, particularly regarding its electric supply systems, along with actions taken with allies, will underscore areas for improvement in the US and initiatives to pursue internationally. To carry out these actions, strong public-private partnerships will be essential.

This document identifies four lines of effort, each of which has direct implications for U.S. military support to allied nations:

Two proposed actions concern the Ukrainian grid: hardening its control network and reconfiguring it to a decentralized topology. These two actions are separate but equally impact Ukraine's security. These two actions should also be considered for all the remaining UA Allies.

The US and its militaries have long focused on hardening Ukraine's electric grid control network. This type of support could extend to other allies facing similar threats. In particular, the Baltic and Nordic States should receive support. Additionally, the U.S. military might assist in transitioning the energy grids of allied countries from centralized to decentralized models to enhance resilience against attacks. This is particularly relevant for countries like Ukraine, where the centralized grid is vulnerable to kinetic attacks. The US DoD's experience in deploying microgrids at key military assets within the US would benefit its allies.

A suggested action to support the use of drones developed in Ukraine is to address weak supply chains for advanced drone components. The US Military could provide substantial logistical and

materiel support to help Ukraine establish robust, diverse supply chains. The US military could also help allied countries develop and integrate advanced drone and AI technologies for surveillance, reconnaissance, and combat operations. The US drones are considered among the best in the world, but the price point for them places them out of consideration for many of its allies.

Finally, the emergence of the cyber-IT army within Ukraine is significant. The document discusses the formation of Ukraine's cyber-IT army and the challenges of integrating civilian volunteers into military operations. It suggests that the U.S. military could assist in coordinating and legitimizing volunteer cyber armies in allied countries.

Appendix A – Key Remaining Tasks

The following tasks are consolidated from the individual sections for ease of reference and tracking. In subsequent versions of this document, these tasks should be assigned ownership, target timelines, and relative priority to support implementation planning.

Section A: Remaining Task

- Assess the possible move of the present Ukrainian Technology Network to a software-defined networking topology.
- Move to a diverse technology blend of hardware and software suppliers in the Technology Network. It must be assumed that adversaries will map the entire IP architecture of the Technology Network, and that hardware-level equipment swaps will increase their costs.
- Inform the Nordic, Baltic, and European States of this Ukrainian methodology and encourage them to replicate appropriate parts. Although more difficult to manage, this concept of using a dynamic network addressing scheme applies directly to other nations of interest.
- Using US military services to support this network management is a reasonable additional burden for the US military.

Note: Given today's evolving political dynamics, NATO and EU member states should seek support from private-sector companies such as ESET.

Section B: Remaining Task

- Plan the gradual transition from a centralized to a decentralized structure based on a replacement strategy.
- Establish the necessary military logistical support groups to develop the required planning.
- Prepare the required DCEI and CCEI list and prioritize
 - Establish the Critical Load Paths for the prioritized DCEI and CCEI facilities.
 - Develop work packages for the top ten DCEI and CCEI Load Paths that address the equipment needs and logistics for the planned modifications to the generation sources and load paths. This work mirrors the approach used by various energy companies to build new energy networks.

Section C: Remaining tasks

- Develop strong NATO-based support for Ukraine's internal drone production

- Western allies should work together to establish secure supply chains for the various materials that Ukraine's internal drone production requires.
- Any efforts the Western allies exert in this regard will be useful in helping the allies produce their drones.
- Develop a strong policy allowing drones to be engaged by the citizenry
 - Much of Western allied countries' critical infrastructure is in the private sector's hands. As such, the security of the infrastructure rests first with the private sector, with federal authorities providing backup.
 - In some countries, such as the US, private companies are typically not permitted to take defensive actions against drones should they be surveilling or attacking their facilities. An established policy that allows the private sector to engage attacking drones provides a critical defensive capability.

Section D: Remaining Tasks

- Convene a senior policy writers' group to draft an EU Cyber IT Army policy allowing citizens to support EU militaries in times of crisis.